

HPS-GPR v5.5.2

A 2015 extension to 100–130 MeV: testing a positive 120 MeV component

11 September 2026 | Study extension of v5.5.1 and v5.0.4

A positive fit is possible, with substantial support dependence

At a fixed mass of 120 MeV, the extended 2015 fit gives a positive Gaussian component of 267.7 ± 163.7 events inside its moving fit window. The error is a local curvature estimate. Its signed fit response is $r = 1.638$; the corresponding 2016 and 2021 values are 2.813 and 0.674. Thus all three standalone amplitudes are positive, although the 2021 component is weak. A 2015 scan from 100 to 130 MeV has its strongest positive interior point at 126 MeV, where $r = 3.803$.

The previous $E^{-2.904}$ residual-response option reduces the three-dataset rate penalty at 120 MeV from 8.992 to 2.055. A refitted exponent -3.543 reduces it to 1.027. This makes the requested comparison possible; it does not establish a shared resonance or a physical beam-energy law.

How the 2015 extension is defined

The saved native spectrum reaches 150 MeV in 0.05 MeV bins. Summing groups of five reproduces the archived 0.25 MeV counts exactly on the original 14–135 MeV support. For this study, the primary training support is 14–150 MeV, chosen to use the available spectrum and preserve upper sidebands across the requested scan. Endpoints at 135 and 145 MeV are retained as support checks wherever at least three bins remain on each side of the signal window. They are not selected by fit strength.

At 120 MeV, the inherited resolution is $\sigma_m = 6.294$ MeV and the moving $\pm 2.25\sigma_m$ window spans 105.84–134.16 MeV. Consequently, the old 135 MeV endpoint leaves only three upper-sideband bins. Extending that support changes the background prediction and the fitted amplitude substantially:

2015 endpoint	Upper bins	$\hat{\epsilon}_{\text{equiv}}^2 [10^{-6}]$	Signed r
135 MeV	3	325.18 ± 105.04	3.103
145 MeV	43	168.35 ± 92.83	1.816
150 MeV	63	138.53 ± 84.74	1.638

The normalization shown as $\epsilon_{\text{equiv}}^2$ is a fitted count amplitude expressed using the inherited prompt-vector conversion. The 90 MeV kernel parameters and effective radiative fraction are held fixed; the inherited linear resolution is extrapolated. No calibration of those assumptions above 90 MeV is added. The 2016 and 2021 spectra, support and conversion are inherited unchanged.

What the fit response means

For each mass and support, $r = \text{sgn}(\hat{a}) \sqrt{2(\text{NLL}_0 - \text{NLL}_{\text{signed}})}$ compares a signed standalone amplitude with zero signal, profiling the same Gaussian-process (GP) background constraint in both fits. It is a conditional local diagnostic, not a calibrated discovery significance. The mass scan uses 0.5 MeV steps; neighboring windows share events and background information. Original release results are unchanged.

The requested mass region and its sidebands

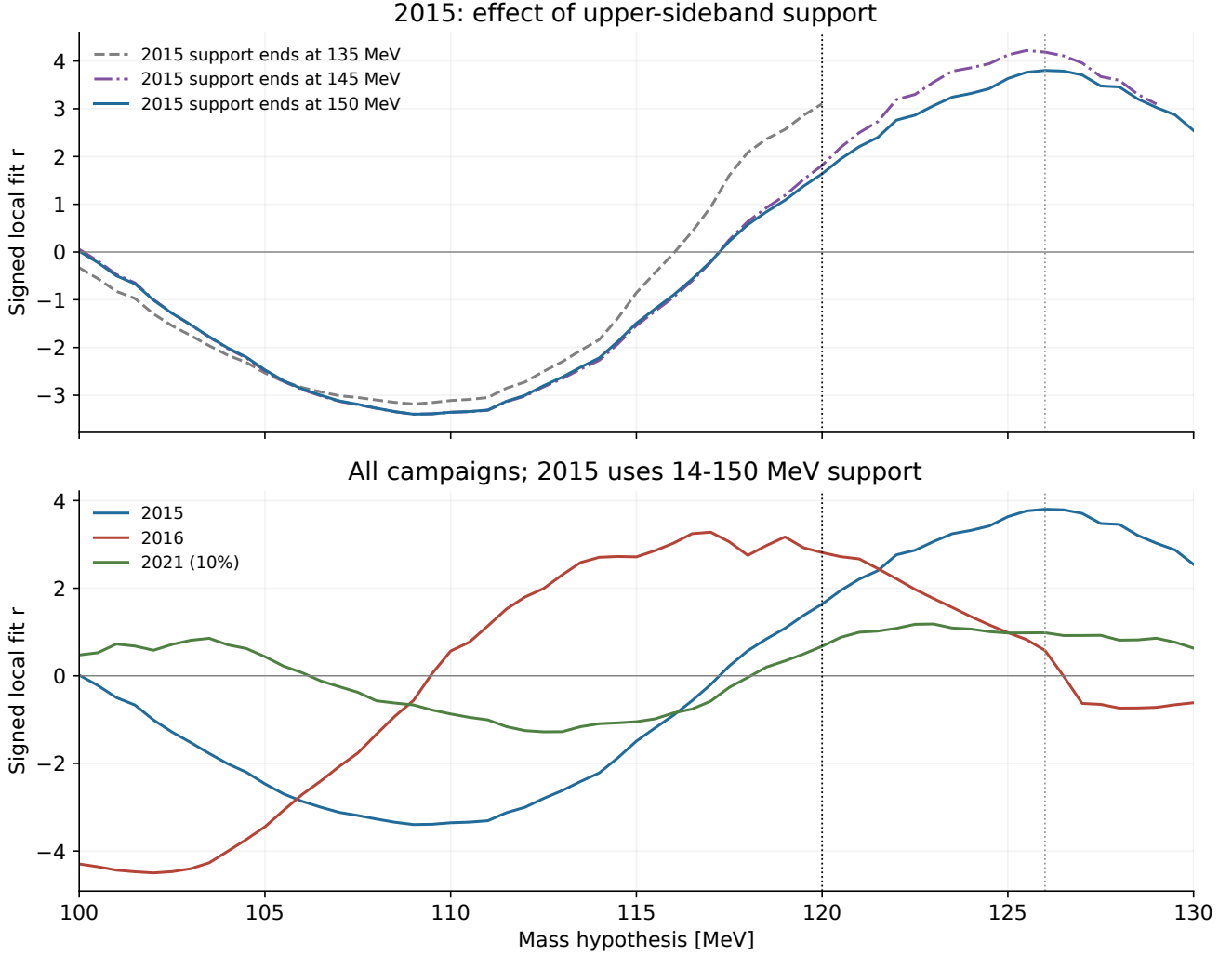


Figure 1: Signed local fit responses. Top: 2015 fits with three training endpoints. The 135 MeV curve stops at 120 MeV because higher hypotheses fail the minimum upper-sideband requirement. Bottom: the three campaigns using the primary 2015 support. Black and gray vertical lines mark the fixed 120 MeV hypothesis and the retrospectively selected 126 MeV point. The plotted roots have no newly calibrated significance interpretation.

The 145 and 150 MeV support choices give similar broad shapes, with positive response developing around 118 MeV and peaking near 126 MeV. At 120 MeV, widening the endpoint from 135 to 150 MeV lowers r from 3.103 to 1.638. At 126 MeV, the 145 and 150 MeV endpoints give 4.185 and 3.803. This sensitivity is part of the result; the larger response is not used to choose the support.

The strongest 2015 point does not coincide with a strong positive response in the other campaigns. At 126 MeV their roots are only 0.579 and 0.983. At the neighboring 121 MeV hypothesis they are 2.669 and 0.995, alongside 2.207 in 2015. These overlapping fits show how the comparison changes with mass; their likelihood values must not be pooled or ranked as a common mass fit.

What is fitted at 120 MeV

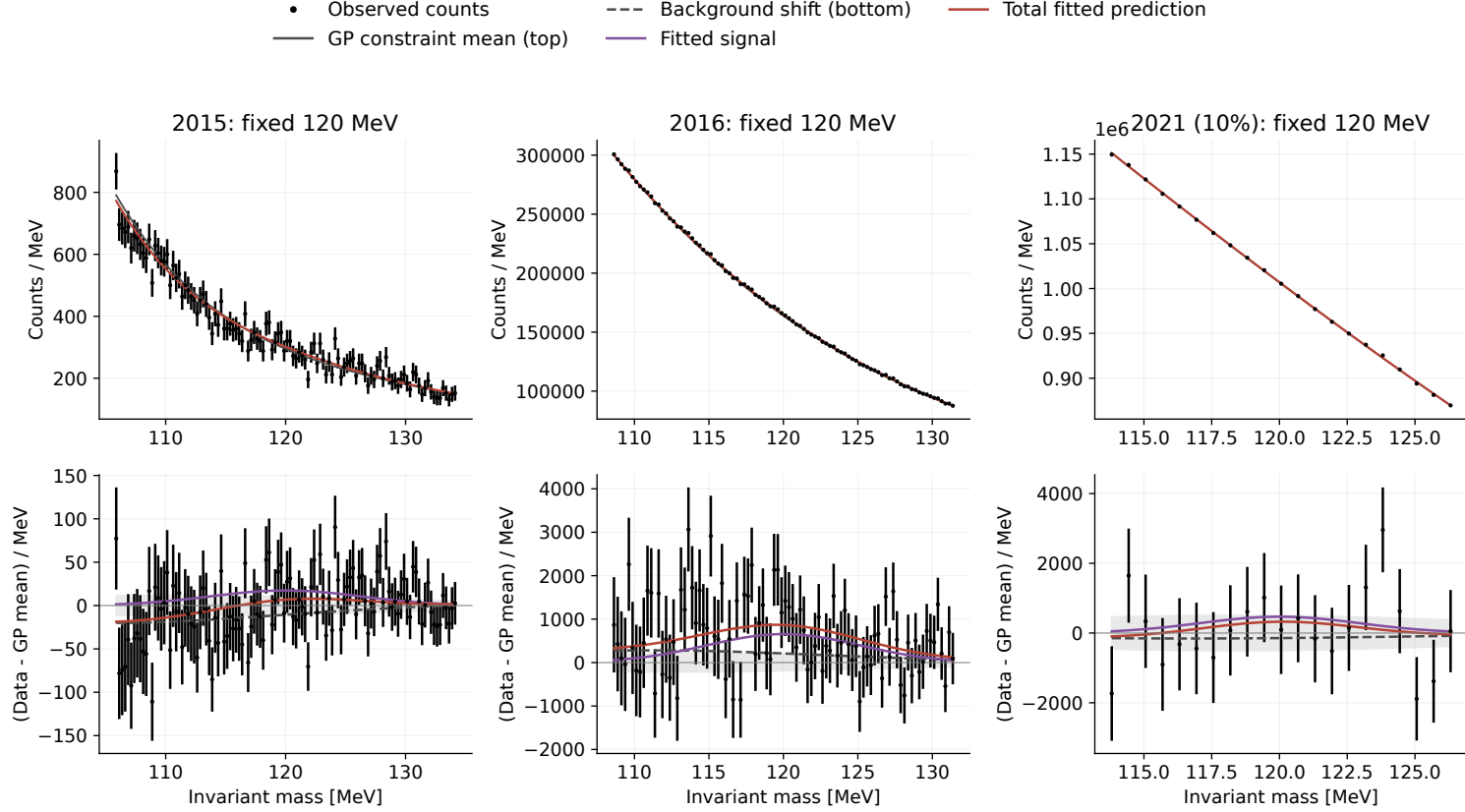


Figure 2: Each campaign has its own fitted amplitude at the same 120 MeV mass, with a profiled GP background. The upper row shows counts per MeV within each moving fit window. The lower row subtracts the GP constraint mean and separates the fitted signal, the background adjustment and their sum. Count bars are $\sqrt{N}$ divided by bin width. Gray shading is the pointwise GP constraint standard deviation, not the uncertainty of the final fitted prediction. All samples and bin widths are inherited, including the 2021 10% sample.

Dataset	$\hat{\epsilon}_{\text{equiv}}^2 [10^{-6}]$	Signed r
2015	138.534 ± 84.736	1.638
2016	13.789 ± 4.904	2.813
2021	0.927 ± 1.374	0.674

These equivalent amplitudes organize a rate comparison. Their 2015 normalization depends on the extrapolated conversion and upper sidebands.

Transferring and refitting the energy response at 120 MeV

The count model is $\lambda_i = b_i + L_i\theta_i + \mu K_i(m)(E_i/2.30 \text{ GeV})^\beta$, with $\mu \geq 0$. Here b_i and $L_i\theta_i$ describe the GP background and its constrained adjustment, and K_i is the inherited count conversion. Beam energies are 1.056, 2.30 and 3.74 GeV [1]. The extra power describes a residual response after the existing normalization [2, 3].

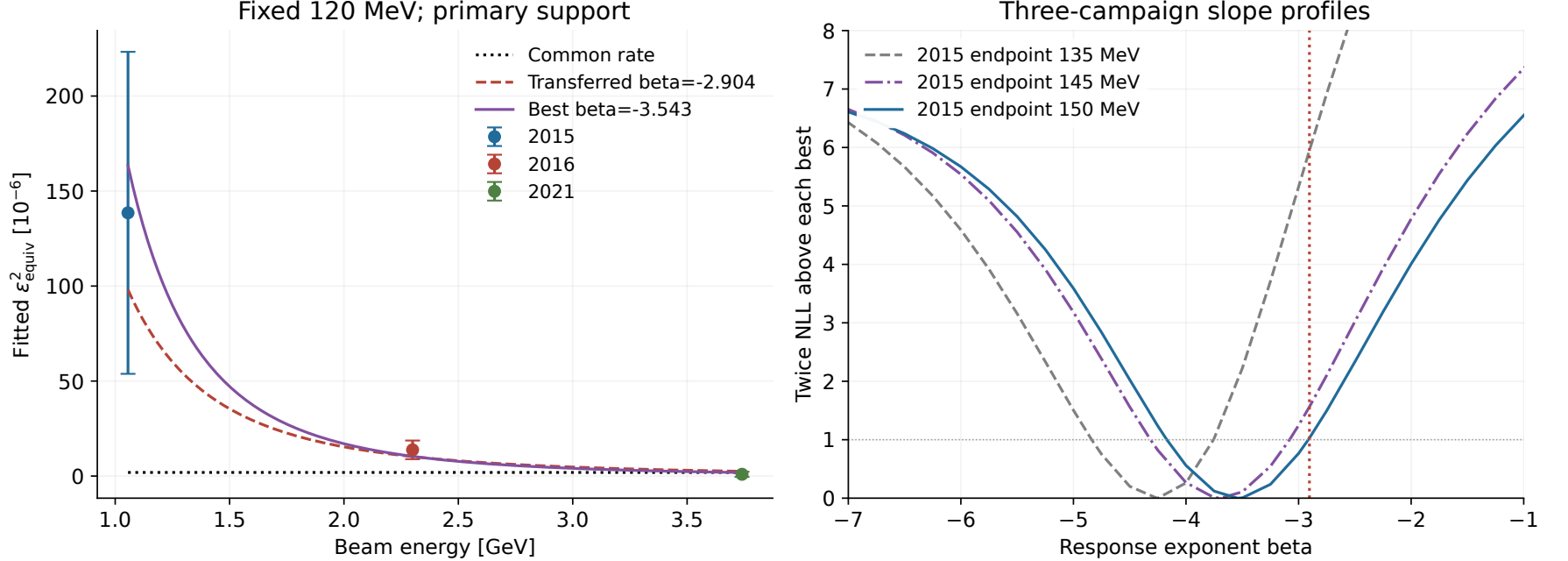


Figure 3: Left: standalone equivalent amplitudes with curvature errors, and common, transferred and refitted response curves. Each curve refits its common amplitude. Right: three-campaign slope profiles for the three 2015 support endpoints, each referred to its own minimum. The dotted vertical line is the frozen $\beta_{92} = -2.90358457$; the horizontal line marks unit twice-NLL change. Curves from different supports do not share an absolute likelihood comparison.

Datasets	$\hat{\beta}$	D_0	D_{92}	D_{best}
2015 + 2016	-2.964	2.168	0.004	0.000
2015 + 2021	-3.960	2.647	1.033	0.000
2016 + 2021	-5.554	6.381	1.787	0.000
2015 + 2016 + 2021	-3.543	8.992	2.055	1.027

$D = 2[\text{NLL}_{\text{model}} - \sum_i \text{NLL}_{i, a_i \geq 0}]$ compares identical local data and constraints. Subscripts 0, 92 and “best” denote $\beta = 0$, the frozen 92 MeV exponent, and a free exponent. Two positive rates can be reproduced exactly by two parameters, so zero pair penalties are automatic. For all three campaigns the best slope is -3.543 , with a connected unit-deviance span $[-4.170, -2.914]$ on $[-12, 12]$; this is a profile-shape diagnostic, not a calibrated interval.

Other hypotheses and pair extrapolations

The same all-three comparison at the inherited 117 and 121 MeV hypotheses, the requested 120 MeV hypothesis, and the selected 2015 maximum gives:

Mass	r_{2015}	r_{2016}	r_{2021}	$\hat{\beta}$	D_{92}	D_{best}
117	-0.200	3.279	-0.579	-3.104	7.293	7.227
120	1.638	2.813	0.674	-3.543	2.055	1.027
121	2.207	2.669	0.995	-3.704	2.083	0.291
126	3.803	0.579	0.983	-5.638	10.387	0.662

Masses are in MeV and all rows use the primary 2015 support. At 117 MeV the negative 2015 and 2021 components resist a common positive rate: even the best response leaves $D = 7.227$. At 121 MeV the three positive rates give $\hat{\beta} = -3.704$ and $D = 0.291$. At 126 MeV the free slope steepens to -5.638 , while transferring β_{92} leaves $D = 10.387$. Its unit-deviance span reaches the lower search bound of -12 ; the very negative tail is weakly constrained. No stable universal response follows from these four overlapping comparisons.

All three pair combinations were fitted at each of the four masses. A pair fits two positive rates exactly, but its extrapolation can be poor. At 120 MeV, fixing the 2016+2021 pair fit predicts 1040.04×10^{-6} in 2015, versus the standalone $(138.53 \pm 84.74) \times 10^{-6}$. Profiling only the unused background gives a plug-in penalty of 110.286. This calculation does not propagate the pair-parameter uncertainty; it is not a rejection probability. Refitting all three datasets allows those parameters to move and gives the much smaller 1.027 penalty.

The other pair predictions at 120 MeV give plug-in penalties 2.892 (unused 2021) and 2.300 (unused 2016). All three have positive fitted backgrounds. Complete results accompany the note; overlapping pair and mass likelihoods are not added.

Interpretation and reproducibility

The requested positive 120 MeV fit exists and admits a rate pattern broadly similar to the original residual-response option. Its magnitude depends on the 2015 upper sidebands, while the stronger 126 MeV component requires a different and weakly constrained slope. Beam energy also changes with campaign, target, detector and selection [1]; the fitted power cannot distinguish those effects. A kinetically mixed vector supplies the benchmark production and normalization framework [3, 2], not a prediction of these empirical exponents. This extension contains 61 scan nodes, 283 standalone fits and 18 slope comparisons. It verifies exact count rebinning, replays the inherited 100 MeV roots within 5.1×10^{-6} , and checks numerical convergence and positive expected counts for the reported models. The full 14–150 MeV spectrum retains at least 19 upper-sideband bins throughout the scan. Numerical checks do not validate the extrapolated kernel, resolution or radiative fraction, the inherited 2016 background qualifications, or retrospective mass selection. No new toys, limits or significance calibration were produced.

Run `scripts/build.sh` to reproduce the fits, figures and note. The package includes input and solver snapshots, source hashes, result tables, rendered-page checks and a portable TeX build. The 126 MeV hypothesis is recorded as selected from the observed scan.

References

- [1] N. Baltzell et al., *The Heavy Photon Search Experiment*. [arXiv:2203.08324](#), Table I.
- [2] P. H. Adrian et al., *Searching for prompt and long-lived dark photons in electroproduced e^+e^- pairs with HPS*. Phys. Rev. D 108, 012015 (2023). [arXiv:2212.10629](#), Eq. (7).
- [3] J. D. Bjorken, R. Essig, P. Schuster and N. Toro, *New Fixed-Target Experiments to Search for Dark Gauge Forces*. Phys. Rev. D 80, 075018 (2009). [arXiv:0906.0580](#).